Factors Influencing Self-reported Vision-Related Activity Limitation in the Visually Impaired

Daryl R. Tabrett[1,2] *and Keziah Latham*[1,2]

Purpose. The use of patient-reported outcome (PRO) measures to assess self-reported difficulty in visual activities is common in patients with impaired vision. This study determines the visual and psychosocial factors influencing patients' responses to self-report measures, to aid in understanding what is being measured.

Methods. One hundred visually impaired participants completed the Activity Inventory (AI), which assesses self-reported, vision-related activity limitation (VRAL) in the task domains of reading, mobility, visual information, and visual motor tasks. Participants also completed clinical tests of visual function (distance visual acuity and near reading performance both with and without low vision aids [LVAs], contrast sensitivity, visual fields, and depth discrimination), and questionnaires assessing depressive symptoms, social support, adjustment to visual loss, and personality.

Results. Multiple regression analyses identified that an acuity measure (distance or near), and, to a lesser extent, near reading performance without LVAs, visual fields, and contrast sensitivity best explained self-reported VRAL (28%–50% variance explained). Significant psychosocial correlates were depression and adjustment, explaining an additional 6% to 19% unique variance. Dependent on task domain, the parameters assessed explained 59% to 71% of the variance in self-reported VRAL.

Conclusions. Visual function, most notably acuity without LVAs, is the best predictor of self-reported VRAL assessed by the AI. Depression and adjustment to visual loss also significantly influence self-reported VRAL, largely independent of the severity of visual loss and most notably in the less vision-specific tasks. The results suggest that rehabilitation strategies addressing depression and adjustment could improve perceived visual disability. (*Invest Ophthalmol Vis Sci.* 2011;52: 5293–5302) DOI:10.1167/iovs.10-7055

Patient-reported outcome (PRO) measures are of increasing importance in vision science and ophthalmology. Such assessments are used to determine the effect of interventions and to guide rehabilitation in the visually impaired by identifying areas of difficulty.[1,2] A wide range of PRO instruments is available to assess health-related or vision-related quality of life and vision-related difficulty in activities of daily living (ADL) (reviewed in Refs. 3, 4). In this study, we considered the aspect of self-report concerned with the perceived difficulties experienced in ADL as a result of vision loss. To better understand what PRO instruments are assessing, we asked how both visual and nonvisual factors influence self-reported, vision-related activity limitation (VRAL) of the visually impaired.

It has been shown that self-reported VRAL as measured by items in common visual function assessment instruments (or questionnaires), represents a unidimensional construct, or the measurement of a single attribute.[2,5,6] The Activity Inventory (AI) is an adaptive instrument designed to provide an individualized assessment of difficulties for a visually impaired respondent.[6] The AI consists of a hierarchal structure in which specific cognitive and motor visual tasks (e.g., pouring or mixing without spilling) underlie more global goals (e.g., preparing meals). Disabilities (or activity limitations, according to the World Health Organization's International Classification of Functioning [WHO ICF] [7]) occur when an individual reports abnormal difficulties in achieving important goals and difficulties achieving a goal are said to depend on the difficulty experienced in the tasks that underlie each goal.[6]

Although self-reported VRAL represents the functional limitations and resultant disability caused by visual impairment, only a modest relationship has been seen between self-report and clinically measured visual function (previously reviewed in Ref. 8). Approximately 50% of the variance in reported difficulty can be explained by clinical measures of visual function including acuity, contrast sensitivity, and visual fields.[8–12]

One possible explanation of the modest relationship between clinical and self-reported visual function may reflect that most instruments ask respondents to report their difficulties under their usual day-to-day conditions, which would include the use of any compensatory strategies such as low-vision aids (LVAs). However, clinical tests of visual function are usually performed under standardized conditions using only refractive correction (either habitual or best corrected). It is possible that despite a decline in clinical visual function, individuals can fully compensate for the decline by using strategies such as LVAs. These patients are said to exhibit preclinical disability as they are at risk of disability, but do not perceive difficulties due to their use of compensatory strategies.[13] However, it is not clear whether self-reported VRAL best reflects the severity of vision loss (clinical function without aids) or the visual function attained when using LVAs.

Factors other than clinical visual function, represented within the WHO ICF, have also been suggested to influence the self-reported VRAL of the visually impaired.[14,15] Depression has often been associated with worse self-reported function in the visually impaired, sometimes independent of the severity of vision loss as characterized by visual acuity.[16] Better social support has been associated with better ADL function in non–vision-specific research,[17,18] yet greater instrumental support from family members has correlated with worse reported VRAL in the visually impaired.[19] Better adaptation and adjustment to vision loss has been significantly associated with fewer re-

From the [1]Department of Vision and Hearing Sciences and the [2]Vision and Eye Research Unit, Anglia Ruskin University, Cambridge, United Kingdom.

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Corresponding author: Keziah Latham, Anglia Ruskin University, East Road, Cambridge, CB1 1PT, UK; keziah.latham@anglia.ac.uk.

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ported functional limitations in ADL as a consequence of vision loss[19] and with greater improvement in reported VRAL after low vision rehabilitation.[20] Personality may also affect self-reported function. Individuals higher in the personality trait of neuroticism have been found to have significantly worse mobility function[21] and poorer self-reported function in ADL[18] in non–vision-specific research.

The purpose of this study was to characterize the relationship between self-reported VRAL as measured by the AI and not only a range of visual function measures, but also nonvisual psychosocial factors in a sample of adults with visual impairment. The purpose was to assess the relative roles of these factors in determining self-reported VRAL, such that what PROs are measuring can be better understood.

METHODS

Subjects

One hundred people participated in the study, which was performed at Essex County and Clacton and District Hospitals and Anglia Ruskin University Eye Clinic. Participants at each location were approached, after having attended low vision support and rehabilitation sessions, and were included if they had experienced vision loss for more than 6 months that they felt caused restriction in their daily life. Those who were under 18 years of age, were unable to perform verbal evaluations in English, had no perception of light in both eyes, or were cognitively impaired (as determined by the Mini-Mental State Examination[22]) were excluded. Ethical approval was granted by Anglia Ruskin University Research Ethics Committee and NHS Essex Ethics Committee. The study complied with the tenets of the Declaration of Helsinki, and all subjects gave informed consent after the nature and possible consequences of the study were explained.

All study interviews and assessments were performed by the same examiner, a qualified optometrist (DRT), usually on the same day, but occasionally in two appointments within 2 weeks of each other. Participation was at least 2 weeks after patients' routine low vision assessments, which were not conducted by DRT. Patients did not need to have been prescribed low vision aids (LVAs) to participate.

Demographics

A structured face-to-face interview elicited key demographics including age, gender, primary cause of visual impairment, and the length of time since ocular diagnosis of each eye, living arrangements, and education. The presence of any co-morbidity from a list of 12 common medical conditions[23] and details of any prescribed medications were recorded and used to represent general health. Focimetry of any current spectacles was undertaken and recorded.

Self-reported VRAL

Self-reported VRAL was assessed using the AI[6] and the results analyzed at goal level, at task level, and for each functional domain at task level (i.e., reading, mobility, visual motor and visual information). The AI contains a wide range of items specifically chosen to represent the visual abilities of a mixed visually impaired population. Adaptive testing is used so that only the difficulty of goals rated as having some importance to the respondent are investigated further. This helps to reduce administration time and produces a tailored set of items for each individual.

Some goals of the original AI that were less relevant in the United Kingdom and deemed less essential in the current sample were not administered (e.g., hunting and shooting). The instrument implemented contained 30 goals and 235 tasks across three objectives (Table 1). The importance of each goal was rated on a four-point scale (not important, slightly important, moderately important, and very important), and the difficulty of important goals and the tasks underpinning these goals were then graded by the respondent on a five-point scale (not difficult, slightly difficult, moderately difficult, very difficult, and impossible without assistance). The typical AI administration time was approximately 35 minutes.

TABLE 1. The 30 Goals of the AI, Nested under Their Associated Objectives

Daily Living
Take care of yourself, such as shave, trim nails, make-up
Take care of health needs
Eat meals
Prepare meals
Perform everyday household tasks such as cleaning, laundry
Recognize people, see expressions and make eye contact
Correspond with others such as mail and letters
Follow news and keep up with current events
Read the time and follow a schedule
Pay bills, manage personal or household finances
Shop for food, clothes, and other necessities
Use the telephone
Choose clothes and dress
Social Interactions
Attend parties and other functions
Entertain guests
Cook or bake for social occasions
Dine out
Attend meetings, such as a club, church, or civic group
Recreational
Provide yourself with leisure entertainment
Exercise
Sew or do needlework
Knit or crochet
Do wood working
Paint or draw
Travel
Perform outdoor recreational activities
Garden for pleasure
Use a computer
Read the newspaper
Do photography

Clinical Visual Function

Distance visual acuity was measured binocularly with an externally illuminated Bailey-Lovie chart[24] at 3 m (chart luminance, 95–100 cd/m^2). Acuity was measured on a per letter basis until no letters on a line could be correctly identified.[25,26] A measure of clinical visual function was assessed with any habitual distance spectacle prescription, and a further measure was obtained with any low-vision aid commonly used for distance tasks (if applicable).

Contrast sensitivity was measured binocularly with a Pelli-Robson Chart[27] at 1 m (chart luminance 95–110 cd/m^2). Participants wore any habitual distance spectacle prescription with a +1.00-D addition where necessary. Contrast sensitivity was measured per letter, until no letters in a triplet could be correctly identified.[28]

Near reading performance was measured binocularly with an MNRead Chart.[29] Reading performance was assessed binocularly at 40 cm with a static reading stand with participants wearing any habitual distance spectacle prescription and a corresponding +2.50-D addition where necessary. If the largest text at this distance could not be seen, the chart was moved closer in a log unit progression until the participant was able to read at least the largest text and the reading addition was altered appropriately. The chart luminance was between 110 and 150 cd/m^2. Text size was expressed in terms of its angular size (logMAR).

A further assessment of reading function with LVAs was made using a different version of the MNRead chart. Any habitual spectacle correction and LVA commonly worn for reading tasks was used. Chart

luminance varied throughout the evaluation (e.g., with the use of illuminated LVAs and variable working distances) depending on participant preference. Since the working distance varied during the assessment, altering the angular degree of the text subtended at the eye, logMAR notation was inappropriate. The physical print size accessed was instead noted in terms of x-height (in centimeters).[30]

For both magnifier-aided and unaided measures, near word acuity, maximum reading speed (MRS), and critical print size (CPS) were determined according to the MNRead manufacturer's instructions.[31] As the MRS is unable to differentiate between participants reading text at different sizes and the CPS is unable to distinguish between participants reading at different speeds, the reading index was calculated.[11] The reading index, which represents MRS as a function of CPS, has been seen to predict self-reported VRAL better than speed and print size separately.[11]

Static threshold binocular central visual fields were conducted with a visual field perimeter (Humphrey Field Analyzer using a Central 30-2 SITA-Fast strategy; Carl Zeiss Meditec, Oberkochen, Germany).[32,33] Participants wore any habitual distance spectacle prescription with a +3.00-D addition where necessary. If the participant could not see the central fixation target, the small then large diamond targets were presented from the test menu and selected if subjectively preferred.[34]

Fixation was manually monitored via the instrument camera view and graded: Good corresponded to the equivalent of less than 30% fixation losses (FL); moderate, <50% FL; and poor, >50% FL. False positives and false negatives were also reviewed as determinants of test reliability. The mean thresholds[33,35,36] of the central 10° and outer 10° to 30° were calculated and used for analysis, to represent macular and nonmacular function, respectively.

Depth discrimination was assessed using the Frisby Stereotest.[37] Participants wore any habitual distance spectacle prescription with an appropriate reading addition corresponding to the test distance where necessary. Depth discrimination tested potentially ranged from 600 to 40 min arc.

Psychosocial Function

Depressive symptoms were assessed with the 15-item Geriatric Depression Scale (GDS).[38] Questions are answered on a dichotomous response scale of yes or no.

Social support was assessed using the 12-item Interpersonal Support Evaluation List (ISEL).[39] The ISEL is designed to assess perceived available support in three areas: appraisal (four statements), belonging (four statements), and tangible support (four statements). Each item is rated on a four-point response scale of definitely true, probably true, probably false, and definitely false.

Adjustment to vision loss was assessed using the 19-item Acceptance and Self-Worth Adjustment Scale (AS-WAS).[40] The AS-WAS evaluates the aspects of adjustment concerned with self-esteem, attitudes, locus of control, self-efficacy, and acceptance. Items are assessed on a four-point response scale of strongly agree, agree, disagree, and strongly disagree.

Personality was assessed according to the five-factor model using the Neuroticism, Extraversion, Openness Five Factor Inventory (NEO FFI).[41] The NEO FFI contains 60 statements relating to the traits of neuroticism, extraversion, agreeableness, conscientiousness, and openness to experience. Each item is rated on a five-point response scale of strongly disagree, disagree, neutral, agree, and strongly agree.

Statistical Analysis

All data were double entered and corrected for errors before data analysis. Rasch analysis[42] was conducted using Winsteps version 3.69.1[43] on responses to all self-report instruments to provide person measure estimates for each construct. Person measures were derived from the AI for overall self-reported VRAL at goal and task level and for each functional domain.

Univariate analyses were first undertaken to explore the demographic, visual function, and psychosocial variables. Bivariate analyses in the form of linear correlation coefficients were then conducted to investigate the association between the predictor variables and self-

Table 2. Descriptive statistics for demographic variables

Sex, *n*/%	39 male, 61 female	
Age, y	81 (73–86)	
Primary ocular diagnosis, *n*/%	Right eye	Left eye
Macular dysfunction, including AMD	54	57
Optic neuropathy, including glaucoma	11	11
Diabetic retinopathy	7	8
Retinal, including myopic degeneration	5	5
Undiagnosed	8	9
Cataract	2	2
Corneal	3	2
Other	10	6
Time since most recent 1° diagnosis, y	5 (3–12)	
Habitual best vision sphere, D	Right eye	Left eye
	0.00 (−0.50 to +1.22)	0.00 (−0.72 to +1.44)
Education, *n*/%		
Standard school leaving age	59	
A-level or equivalent	13	
Further education	12	
Higher education	2	
Bachelors degree	10	
Post graduate qualification	4	
Living arrangements, *n*/%		
Alone	40	
With partner	51	
With other	6	
Warden assisted	3	
Number of prescribed medications	4 (2–6.75)	
Number of co-morbidities	1.5 (1–3)	

$n = 100$. As the distribution of data was significantly different from normal, the median (25%–75% interquartiles) values are given.

Table 3. Details of Aids Chosen and Used by Participants for Distance and Near Visual Function Assessments with LVAs

LVA	(%)
Distance	
Telescope	29
None	71
Near	
Illuminated hand magnifier	31
Illuminated stand magnifier	28
Non-illuminated hand magnifier	16
Reading spectacles	9
Illuminated flat-field magnifier*	8
Other (e.g. reduced working distance, additional light)	8

$n = 100$.
* Makrolux; Eschenbach Optik of America, Danbury, CT.

reported VRAL. Next, significantly correlated variables were entered into separate demographic, visual function, and psychosocial factor stepwise regression analyses. Finally, the variables identified as significant predictors were entered into an overall multiple regression model. Conducting separate regression analyses initially and then together in overall regression analyses allowed the amount of unique variance of self-reported VRAL explained by visual and nonvisual factors to be calculated in addition to determining the degree of shared variance they explain.

Results

Table 2 summarizes the descriptive characteristics of the study population. Table 3 provides details of the aids used by participants for the assessment of magnifier-aided visual function. Details of the self-reported and visual function measures and psychosocial factors within the sample are given in Table 4. As only eight participants had measurable depth discrimination, participants were said either to be able to achieve a positive result on the Frisby Stereotest or not to be able, which was used in further analyses. For binocular visual field assessment, similar to automatically determined monocular fixation distributions,[44] fixation was rated as good in 64% of participants and 98% were adjudged to have at least moderate fixation. Ninety-seven percent of false-positive statistics were 20% or less and 99% were 25% or less, and 88% of false negatives were 20% or less and 91% were 25% or less. For the self-report instruments, more positive logit values correspond to greater amounts of the construct measured, and greater visual ability in regards to the AI.

Further details of the Rasch analyses of the psychosocial instruments are given in Table 5. The MNSQ infit and outfit statistics indicated broadly acceptable item fit for all scales (where values between 0.60 and 1.40 are deemed acceptable).[45] Principal component analysis of the residuals identified sufficient unidimensionality (first contrast eigenvalues between 1.8 and 2.3, where eigenvalues up to between 2.0[46] and

Table 4. Descriptive Statistics of the Self-reported and Clinical Visual Function Assessments

	Average	Min–Max
Self-reported VRAL		
AI goals person measure (logits)	1.98 ± 1.74	−2.96–8.14
AI all tasks person measure (logits)	1.30 ± 1.23	−2.31–5.20
AI reading person measure (logits)	0.60 ± 1.70	−4.45–5.73
AI mobility person measure (logits)	2.44 ± 1.84	−2.47–6.70
AI visual motor person measure (logits)	1.76 ± 1.33	−1.88–5.77
AI visual information person measure (logits)	1.69 ± 1.56	−2.77–7.59
Clinical Visual Function		
Distance visual acuity, logMAR	0.84 ± 0.36	0.08–1.64
Distance visual acuity with LVA, logMAR*	0.59 ± 0.40	−0.20–1.36
Contrast sensitivity, logCS units	1.10 (0.88–1.35)	0.15–1.75
Reading acuity, logMAR	0.81 ± 0.45	−0.04–2.00
CPS, logMAR	1.00 (0.80–1.30)	0.10–2.00
MRS, wpm	107.36 (45.95–143.7)	1.34–210.52
Reading index, MRS/CPS	105.62 (35.76–190.31)	0.67–1758.50
Reading acuity with LVA, *x*-height, cm	0.077 (0.056–0.15)	0.024–1.27
CPS with LVA, *x*-height, cm	0.18 (0.12–0.35)	0.05–1.16
MRS with LVA, wpm	100.84 (43.41–142.52)	3.10–200.67
Reading index with LVA, MRS/CPS	510.64 (122.55–1089.20)	2.67–4013.40
Central 30–2 mean threshold, dB	22.28 (14.21–26.40)	0.11–30.32
Central 10 mean threshold, dB	22.91 (14.06–27.31)	0.50–34.25
Central 10 to 30 degree mean threshold, dB	21.58 (13.04–26.31)	0.00–29.27
Depth discrimination, sec arc		
Yes, $n = 8$	396.88 ± 192.63	55–600
No, $n = 92$	—	—
Psychosocial Factors		
Depression, GDS, logits	−2.40 (−3.36–−1.26)	−4.77–3.15
Social support, ISEL, logits	1.28 ± 1.07	−1.37–4.96
Adjustment to vision loss, AS-WAS, logits	1.53 (0.84–1.93)	0.71–4.69
Personality, NEO FFI, logits		
Neuroticism	−0.84 (−1.33–−0.25)	−5.15–2.80
Extraversion	0.33 ± 0.67	−1.05–1.93
Openness	0.05 ± 0.52	−1.43–1.30
Agreeableness	1.39 ± 0.73	−0.25–4.31
Conscientiousness	1.23 (0.70–2.19)	−0.44–4.72

$n = 100$. The mean ± SD is provided for the variables whose frequency distributions were not significantly different from normal. For all others, the median (25%–75% interquartiles) is given.
* $n = 29$.

TABLE 5. Psychometric Properties of the GDS, ISEL, AS-WAS, and NEO FFI Following Rasch Analysis

	Infit MNSQ		Outfit MNSQ						Variance Explained (%)	
	Mean ± SD	Range	Mean ± SD	Range	Targeting	Person Separation	Item Separation	1st contrast Eigenvalue	Empirical	Modelled
GDS	0.98 ± 0.19	1.39–0.77	0.99 ± 0.38	1.78–0.47	−2.09	1.11	3.48	1.8	35.3	33.6
ISEL	1.01 ± 0.14	1.32–0.84	0.98 ± 0.15	1.25–0.86	1.28	1.93	2.80	2.1	38.3	38.3
AS-WAS	1.01 ± 0.11	1.25–0.76	0.99 ± 0.15	1.29–0.72	1.49	2.38	5.32	2.0	46.7	47.4
NEO FFI										
Neuroticism	1.02 ± 0.22	1.33–0.61	1.01 ± 0.20	1.37–0.64	−0.91	2.61	3.24	2.1	50.1	50.2
Extraversion	0.99 ± 0.16	1.25–0.78	0.99 ± 0.16	1.34–0.81	0.33	1.74	7.15	2.3	48.2	44.7
Openness	0.99 ± 0.32	1.57–0.56	1.03 ± 0.34	1.68–0.59	0.05	1.68	4.95	2.2	35.2	33.4
Agreeableness	1.02 ± 0.27	1.67–0.48	0.99 ± 0.24	1.45–0.52	1.39	1.45	6.74	2.0	46.5	47.7
Conscientiousness	0.99 ± 0.25	1.33–0.61	1.02 ± 0.25	1.42–0.63	1.49	2.21	3.88	2.0	39.8	40.4

Mean ± SD are given for normally distributed person measures and the median (interquartile range) provided when significantly different to normal.

3.0[47] have been used previously to indicate unidimensionality). Person separation was greater than 2.0 for the AS-WAS, neuroticism, and conscientiousness scales, a value that indicates sufficiently reliable person ordering,[1] but was less than 2.0 for all other psychosocial scales. Notably, the GDS had a less than optimal person separation of 1.1, most likely due to the few items targeted toward the less depressed[48]: suboptimal person separations have previously been noted with depression scales (Ahmadian L, et al. *IOVS* 2008;49:ARVO E-Abstract3755).[49] Item separation was greater than 2.0 for all scales, indicating reliable item ordering. The psychosocial scale measures determined were used in further analyses as shown, with no adaptations made to adjust the psychometric properties of the scales on the basis of the Rasch analysis.

Table 6 shows the variables significantly correlated with self-reported VRAL in bivariate analyses (Spearman's *r*). Because of the multiple number of comparisons performed ($n = 156$), a Bonferroni correction would result in a significance level of $P = 0.00032$ (0.05/156) at the 5% level. However, because of the exploratory nature of the analyses and in accordance with others,[8] the corrected *P* value could be considered excessively stringent.[50] In its place, it should be noted that approximately eight (156×0.05) correlations would be expected to be significant by chance alone at the $P = 0.05$ level. Depth discrimination, personality (except neuroticism), and demographic variables (except age with self-reported reading function) were not significantly associated with any domain of the AI.

The visual functions and psychosocial variables significantly associated with self-reported VRAL, along with age, were entered into separate stepwise regression analyses (Table 7). For visual function, magnifier-aided distance visual acuity was not included as a predictor variable, as it correlated strongly with visual acuity without magnifiers ($r = 0.92$, $P < 0.001$), risking multicollinearity, and was underrepresented in the sample ($n = 29$, see Table 4).

Table 7 shows that in most domains, either near word (RA) or distance letter (VA) acuity, and reading index (RI), along with visual field function of the central 10° to 30° (OR), best explained the self-reported VRAL. For self-reported reading function, contrast sensitivity (CS) replaced visual fields as a significant predictor, and contrast sensitivity along with the central 10° to 30° field best explained mobility function. With regard to nonvisual psychosocial factors, depression (GDS) and/or adjustment (AS-WAS) were significant predictors of self-reported VRAL. Social support and neuroticism lost significance when considered alongside depression and adjustment. Age did not remain significantly associated with self-reported function.

Overall multiple regression analyses (Table 8) were conducted which included the significant predictors determined in the above regression analyses. The total variance in self-reported VRAL explained by both visual function and psychosocial factors was between 59% and 71%. For each regression model, the residuals were normally distributed and the assumptions of homoscedasticity (the extent to which the variance of residuals was equal for all predicted values) and linearity were supported. The variation inflation factor was sufficiently low for all variables indicating absence of bias from multicollinearity. The standardized residuals supported adequate fit of the sample to the model.

TABLE 6. Bivariate Analyses between Clinical Visual Function and Psychosocial Factors as Compared with Self-reported VRAL

	VA	MVA	CS	RA	RI	MRA	MRI	C10	OR	Depression	Adjustment	Neuroticism	Social Support	Age
Goals	−0.60	−0.39	0.53	−0.64	0.62	−0.49	0.60	0.46	0.37	−0.52	0.56	−0.40	0.28*	-0.09^{ns}
All tasks	−0.66	−0.47	0.59	−0.69	0.67	−0.52	0.63	0.51	0.42	−0.50	0.51	−0.41	0.22*	-0.08^{ns}
Reading	−0.71	−0.56	0.57	−0.75	0.72	−0.63	0.71	0.44	0.33	−0.40	0.42	−0.26	0.15^{ns}	−0.21*
Mobility	−0.27	-0.08^{ns}	0.47	−0.30	0.33	-0.18^{ns}	0.29	0.49	0.61	−0.50	0.52	−0.55	0.31	0.13^{ns}
Visual motor	−0.56	−0.44	0.42	−0.59	0.55	−0.43	0.50	0.38	0.31	−0.51	0.50	−0.43	0.25*	-0.03^{ns}
Visual information	−0.61	−0.45	0.55	−0.64	0.62	−0.43	0.56	0.53	0.42	−0.49	0.47	−0.40	0.20*	-0.07^{ns}

Nonparametric two-tailed Spearman's correlation coefficients (*r*) are provided due to many variables being significantly differently distributed from normal. VA, distance visual acuity; MVA, magnifier-aided distance visual acuity; CS, contrast sensitivity; RA, near reading acuity; RI, reading index; MRA, magnifier-aided near reading acuity; MRI, magnifier-aided reading index; C10, central 10° mean threshold; OR, outer ring mean threshold (10°–30°).

^{ns} $P > 0.05$.

* $P < 0.05$.

Without symbol: $P < 0.01$.

TABLE 7. Results of Separate Stepwise Regression Analyses for Clinical Visual Function and Psychosocial Factors Predicting Self-reported VRAL

Clinical Visual Function					Psychosocial Factors				
	B	SE B	β	R^2 Change		B	SE B	β	R^2 Change
Goals					Goals				
Constant	2.40	0.55			Constant	0.31	0.28		
VA	−2.03	0.42	−0.42***	0.41***	AS-WAS	0.62	0.18	0.34**	0.26***
RI	0.003	0.001	0.29**	0.07**	GDS	−0.36	0.12	−0.31**	0.07**
OR	0.05	0.016	0.23**	0.05**					
Total R^2				0.53					0.33
All tasks					All tasks				
Constant	1.36	0.34			Constant	0.15	0.20		
RA	−1.21	0.23	−0.45***	0.46***	AS-WAS	0.46	0.13	−0.36**	0.26***
OR	0.036	0.011	0.24**	0.06**	GDS	−0.23	0.08	−0.28**	0.05**
RI	0.002	0.001	0.27**	0.05**					
Total R^2				0.57					0.31
Reading					Reading				
Constant	0.38	0.61			Constant	−0.58	0.28		
RA	−1.73	0.32	−0.46***	0.56***	AS-WAS	0.79	0.16	0.45***	0.20***
CS	1.24	0.39	0.25**	0.05**					
RI	0.002	0.001	0.23**	0.04**					
Total R^2				0.64					0.20
Mobility					Mobility				
Constant	−0.95	0.48			Constant	0.72	0.30		
OR	0.12	0.02	0.54***	0.40***	GDS	−0.46	0.12	−0.38***	0.27***
CS	1.09	0.50	0.20*	0.03*	AS-WAS	0.51	0.19	0.26*	0.05*
Total R^2				0.43					0.32
Visual motor					Visual motor				
Constant	1.94	0.41			Constant	0.53	0.22		
RA	−1.23	0.28	−0.42***	0.37***	AS-WAS	0.46	0.14	0.33**	0.25***
OR	0.033	0.013	0.21*	0.04**	GDS	−0.26	0.09	−0.30**	0.06**
RI	0.001	0.001	0.21*	0.03*					
Total R^2				0.44					0.31
Visual information					Visual information				
Constant	2.30	0.049			Constant	0.33	0.26		
VA	−2.08	0.037	−0.48***	0.43***	GDS	−0.34	0.11	−0.33**	0.23***
OR	0.048	0.014	0.26**	0.07***	AS-WAS	0.43	0.17	0.27*	0.05*
RI	0.002	0.001	0.19*	0.03*					
Total R^2				0.52					0.28

Clinical visual function variables entered: VA, CS, RA, RI, MRA, MRI, C10, and OR (key: see Table 6). Psychosocial factor variables entered: depression (GDS), adjustment (AS-WAS), neuroticism and social support.

* $P < 0.05$.

** $P < 0.01$.

*** $P < 0.001$.

Using results from all regression analyses, the percentage of unique variance of self-reported VRAL explained by visual and nonvisual factors was determined (Table 9). The variance explained by visual function (total R^2 value in Table 7) represents both variance explained solely by visual function (c) and variance shared with psychosocial factors (b). Similarly, the total variance explained by psychosocial factors in Table 7 represents both variance explained solely by psychosocial factors (a) and variance shared with visual function (b). As the total variance explained by all parameters (a + b + c) is known (total R^2 values in Table 8), the unique variance accounted for solely by visual function (c) and uniquely by psychosocial factors (a) can be calculated as, for example: (c) = (a + b + c) − (a + b). Finally, shared variance can be calculated as (b) = (a + b + c) − (a) − (c) and represents the proportion of self-reported VRAL variance explained by both clinical visual function and psychosocial factors.

Clinical visual function could explain between 28% and 50% unique variance of self-reported VRAL depending on functional domain and nonvisual psychosocial factors accounted for between 6% and 19% unique variance. These proportions demonstrate that both visual and nonvisual factors are associated with self-report independent of each other. In addition, due to the relatively constant level of shared variance (10%–17%), the results also suggest that the association of visual and nonvisual factors with self-report is to a certain extent dependent on the other, regardless of the self-reported visual domain. This relationship was therefore further explored (Table 10).

In Table 10, the linear correlation coefficients between the visual and nonvisual factors show that adjustment, but not depression, was significantly associated with the severity of vision loss. Therefore, it can be said that depression is significantly associated with self-reported VRAL independent of vision loss severity, whereas the association of adjustment is partly dependent on the degree of vision loss. As the significant relationship between adjustment and clinical visual function was lost when self-reported VRAL is controlled for, the results indicate that adjustment is only associated with vision loss severity due to the (self-reported) functional limitations vision loss causes rather than the actual degree of visual loss itself.

DISCUSSION

The results of the study indicate that the nonvisual psychosocial factors of depression and adjustment can explain unique variance in self-reported VRAL not accounted for by the severity of vision loss as represented by clinical visual function. The

TABLE 8. Overall Regression Analyses Indicating the Significant Predictors of Self-reported VRAL

	B	SE B	β	R^2 Change		B	SE B	β	R^2 Change
Goals					All tasks				
Constant	1.35	0.47			Constant	0.68	0.30		
VA	−1.81	0.34	−0.37***	0.41***	RA	−1.08	0.19	−0.40***	0.46***
GDS	−0.36	0.80	0.31***	0.17***	AS-WAS	0.22	0.09	0.17*	0.14***
RI	0.002	0.001	0.27***	0.07***	GDS	−0.22	0.06	−0.27***	0.04**
OR	0.03	0.013	0.16*	0.03**	RI	0.002	0.000	0.25***	0.05***
AS-WAS	0.30	0.13	0.16*	0.02*	OR	0.03	0.009	0.18**	0.03**
Total R^2				0.69	Total R^2				0.71
Reading					Mobility				
Constant	−0.07	0.57			Constant	−1.64	0.42		
RA	−1.70	0.29	−0.46***	0.56***	OR	0.10	0.02	0.46***	0.40***
AS-WAS	0.45	0.10	0.26***	0.09***	GDS	−0.51	0.08	−0.42***	0.17***
CS	1.07	0.36	0.21**	0.04**	CS	0.99	0.42	0.18*	0.02*
RI	0.002	0.001	0.19**	0.02**					
Total R^2				0.70	Total R^2				0.60
Visual motor					Visual information				
Constant	1.59	0.31			Constant	1.52	0.44		
RA	−1.18	0.24	−0.40***	0.37***	VA	−1.88	0.32	−0.43***	0.43***
GDS	−0.26	0.07	−0.29***	0.15***	GDS	−0.38	0.06	−0.37***	0.15***
RI	0.001	0.001	0.20*	0.04**	OR	0.04	0.01	0.20**	0.05**
AS-WAS	0.27	0.11	0.20*	0.03*	RI	0.002	0.001	0.21**	0.03**
Total R^2				0.59	Total R^2				0.65

Entered variables are the significant predictors from Table 7, and the key to the variables is given in Table 6

* $P < 0.05$.

** $P < 0.01$.

*** $P < 0.001$.

visual factors that are associated with self-reported difficulties correspond to visual function without, as opposed to with, LVAs. Adjustment, but not depression, is significantly associated with the degree of vision loss but only when self-reported function is not considered.

The specific clinical visual functions best explaining self-reported VRAL are visual acuity and reading performance without LVAs, contrast sensitivity, and a measure of visual fields. These visual functions have all previously been seen to significantly correlate with self-reported function.[8–12] As acuity without LVAs was the strongest predictor of self-reported VRAL in most instances, the findings of previous studies in which only visual acuity is considered to represent vision loss severity can be more strongly regarded.[16]

The AI asks respondents about difficulty with tasks even when wearing glasses and using magnifiers or other LVAs. We hypothesized that visual acuity with LVAs would better predict reported activity limitation than standard clinical measures of acuity. Few participants used LVAs for distance tasks, also seen elsewhere,[51] and so we were able to test this prediction only for reading ability. Contrary to the hypothesis, self-reported VRAL in all domains was better predicted by visual function without LVAs.

It is understandable that reading performance with LVAs may not be a good representative of function in activities other than reading. However, this fails to explain why reading function with LVAs does not correlate well with reported habitual reading performance. As the study appointments followed low vision support with a separation of a minimum of 2 weeks, it is possible that participants were not accustomed to using any newly prescribed aids, affecting their perceived habitual limitations. Allowing a greater period to habituate with any aid may strengthen the relationship between self-reported VRAL and magnifier-aided visual function.

However, all participants regularly used some form of reading aid. In addition, magnifier-aided near reading performance was better than function without aids, suggesting proficiency of use (median x-height values with LVAs [Table 4] correspond to print size of 0.49 logMAR [CPS] and 0.12 logMAR [acuity] viewed at 40 cm, smaller than the equivalent assessments

TABLE 9. Proportions of Variance of Self-reported VRAL Explained by Clinical Visual Function and Psychosocial Factors for Each Domain of the AI

	Psychosocial Variables (a)	Shared Variance (b)	Clinical Visual Function (c)	Total Variance Explained (a+b+c)
Goals	16%	17%	36%	69%
All tasks	14%	17%	40%	71%
Reading	6%	14%	50%	70%
Mobility*	17%	10%	33%	60%
Visual motor†	19%	12%	28%	59%
Visual information*	13%	10%	42%	65%

* Values calculated excluding adjustment as it lost significance in the overall regression models for these domains (see Table 8).

† Values calculated excluding central 10° to 30° degrees as it lost significance in the overall regression model for this domain (see Table 8).

TABLE 10. Bivariate Analyses between Clinical Visual Function and Psychosocial Factors

Control Variables		VA	RA	CS	RI	OR
None	GDS	0.16^{ns}	0.17^{ns}	-0.13^{ns}	-0.07^{ns}	-0.17^{ns}
	AS-WAS	-0.20^{*}	-0.21^{*}	0.23^{*}	0.26^{*}	0.24^{*}
Self-reported VRAL	GDS	-0.16^{ns}	-0.12^{ns}	0.24^{ns}	0.24^{*}	0.20^{ns}
	AS-WAS	0.15^{ns}	0.12^{ns}	-0.10^{ns}	0.10^{ns}	-0.10^{ns}

Zero-order correlation coefficients and partial correlation coefficients where self-reported VRAL is controlled, are provided.

$^{*}P < 0.05$.

$^{ns}P > 0.05$.

without LVAs: 1.00 logMAR and 0.81 logMAR, respectively). Therefore, at least when considering responses to the AI, it may also be that despite the use of aids, difficulty is still reported, with the difficulties reflective of the severity of vision loss as represented by standardized clinical visual function, as opposed to function with LVAs. Further, although all participants were instructed and reminded when required to report their difficulties when using any LVAs such as magnifiers, it is possible that individuals are more often without their magnifiers than using them, and this is what is reflected in self-reported visual function.

It should be noted that in the present study and others,[11,52] clinical visual function was best associated with self-reported VRAL in more vision-specific items, notably reading tasks (50% unique variance explained; Table 9). Reading items include tasks such as reading newspaper articles, which are closely reflected by clinical assessments such as reading performance and as a result there is perhaps less chance of individual interpretation and influence from psychosocial factors (6% unique variance explained). A greater influence of nonvisual psychosocial factors was seen when self-report included fewer vision-specific items, which may be open to greater individual interpretation, such as many of the mobility and visual motor items (up to 17% unique variance explained). Examples of such items include arranging and using transportation (mobility), and listening to the radio and using a computer (both visual motor). The visual requirements for these tasks are at face value less comparable to the clinical assessments of visual function resulting in a weaker relationship with clinical visual function (28%–33% unique variance explained). Therefore, when using PROs of ADLs the choice of questions or instrument should be driven by its intended purpose, depending on whether the aim is to reflect visual function (use vision specific tasks) or to give an overview of perceived disability (use less vision-specific tasks).

With reference to the psychosocial factors, depression has been significantly associated with self-reported VRAL,[16,19,53] explaining up to 20% variance after regression analyses.[54,55] The results of the present study demonstrate that, except for reported reading function, depression explains variance in self-reported responses not accountable for by visual impairment severity, including as much as 17% of variance in self-reported mobility function.

Adjustment to vision loss depends on the psychosocial profile of an individual, although the exact process of adjustment is still largely unknown[40,56] and individuals can vary enormously in their reactions to vision loss.[57,58] Adjustment in the context of the present study refers to the unidimensional latent construct that involves aspects of self-esteem, acceptance, locus of control, attitudes, and self-efficacy, as measured by the 19-item AS-WAS.[40] The results indicate, consistent with previous studies,[19,59] that adjustment to vision loss is significantly associated with self-reported VRAL. The association between adjustment and self-reported limitation is not entirely independent of the severity of vision loss, but is only apparently related to the severity of vision loss due to the functional limitations caused by the visual impairment rather than with the degree of vision loss itself, since the significant relationship between adjustment and clinical visual function is lost when self-reported VRAL is controlled. Therefore, greater vision loss does not necessarily indicate poorer adjustment. Clinically this is important as although greater vision loss may be related to poorer adjustment the effects could be limited by reducing the functional limitations through rehabilitation.

Despite being significantly associated with self-reported VRAL (Table 6), the effect of social support is lost when the levels of depression and adjustment are also considered (Table 8). Together with the findings of others,[60] it appears that social support does not directly influence self-reported activity limitation; rather it is depression and adjustment to which social support is related.[59,61] Similarly, and in agreement with other studies,[62,63] the significance of personality, namely neuroticism, in influencing self-report is also lost when depression and adjustment are considered. However, neuroticism may be related to depression[64,65] which does predict self-reported VRAL.

Although low vision rehabilitation often seeks to improve visual function with the use of optical and adaptive devices,[66] the results of the present study suggest that difficulty in ADL may still be perceived despite implementing compensatory strategies such as LVAs. As vision loss is generally irreversible, and since depression and aspects in the adjustment process such as self-esteem and self-efficacy are dynamic,[67] it may be advisable to also target psychosocial functioning to help alter perceptions of visual function with rehabilitation. Indeed, improvements in depression and adjustment post rehabilitation have paralleled improvements in self-reported VRAL.[68–71] Nonetheless, it appears the next step in research is to study multiple time points, including the impact of different interventions[72] and structural equation modeling techniques,[73] to further investigate the importance of depression, adjustment, and severity of vision loss on self-reported limitations as identified in the present study.

Limitations

Because of ethical considerations, participation in the study was entirely voluntary, which may have led to response bias. However, the ocular diagnoses of the current sample are comparable to the U.K. Blind Register[74] with a large range of vision loss severity represented (Table 4). A wide variation in levels of depression and adjustment was also seen, with a significant number (23%) exhibiting at least mild depressive symptoms (raw score, $\geq$5; Table 4).

Although the psychosocial instruments were chosen as established measures, it should be acknowledged that some of the psychometric properties (namely, person separation) of the instruments were less than optimal. In future, scales with more items spanning a wider ability range are advisable.

Although the AI was originally designed to assess VRALs, it is possible that activity limitations caused by factors other than vision loss may influence responses. However, neither health co-morbidities nor the number of prescribed medications were related to AI person measures, suggesting that when administered face-to-face in a vision rehabilitation environment the responses to the questions do primarily reflect VRALs. The potential impact of non–vision-related limitations is an issue shared with other instruments, which could be minimized by specifically asking whether the difficulties are due to vision loss.[75] The present findings may therefore generalize only to others using the AI in a similar context, and further work is needed to examine whether the findings, such as the lack of relationship between activity limitation and function with LVAs, are specific to the AI or general to all VRAL instruments.

Conclusion

As stated by Freeman et al.[76] it is important to understand the different components that are related to self-reported VRAL to either improve that element or else to teach patients how to cope better with loss of that type. From the results of the present study, self-reported VRAL, as assessed by the AI at the goal and all-task level, and for visual information and visual motor task domains is most influenced by acuity without LVAs, reading index without the use of LVAs, and visual field function of the central 10° to 30°. For reading tasks, contrast sensitivity takes the place of visual fields in importance, and for mobility tasks contrast sensitivity and the visual field function of the central 10° to 30° are relevant, but acuity is not. The psychosocial factors of depression and adjustment to vision loss are also related to all domains of self-reported VRAL, independent of clinical visual function, especially when self-reported VRAL is assessed using less vision-specific items. These findings can be used to guide the interpretation of AI measures and future derivations of self-report instruments.

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